ASUS UGen AI League · Stage I Proposal · Battlefield Lightning · Global Online Track

AegisEdge

A privacy-first, on-device PPE and restricted-zone alert system built on the ASUS UGen300 USB AI Accelerator

Theme: Workplace AI · Team: refrigerator1

Problem: Safety Monitoring Needs Real-Time, Private Vision

Sites rely on helmets and high-vis vests, yet cameras mostly record incidents for later review and manual supervision leaves coverage gaps.

Privacy

Streaming worker video to the cloud raises privacy and union concerns and can expose site information.

Latency

Cloud round-trips add network-dependent delay. A local pipeline has a predictable, measurable delay.

Cost

Cloud GPU subscriptions and uplink bandwidth grow with every camera, a heavy burden for SMEs.

how can a site get real-time PPE alerts without sending video off-site?

Solution: Local Pipeline on Existing Hardware

IP Camera

RTSP streams over the factory LAN.

Host PC

Decodes and pre-processes frames; USB 3.1 Gen2 link to the accelerator.

UGen300

Hailo-10H runs object detection (40 TOPS INT4, 20 TOPS INT8).

Rule Engine

PPE checks and zone geofencing with multi-frame confirmation.

Local Output

Dashboard, event log, relay or Modbus alarm.

Positioned as an
. It supplements, and never replaces, certified safety PLCs and emergency stops.

Methodology: Model, Quantization, Validation

Detection model

- Start from a Hailo Model Zoo YOLOv8n-class model and fine-tune
- Classes: person, helmet, vest (or no_helmet, no_vest, chosen by validation)
- Restricted zones are polygons in the rule engine, not a detection class

Quantization and validation

- ONNX to .hef with the Hailo Dataflow Compiler, INT8 with site-style calibration images
- Report FP32 vs INT8 mAP on a held-out set
- YOLOv8 is AGPL-3.0: evaluate a commercial license or a permissive alternative

Privacy by Design and Known Limits

Privacy by design

- Frames are processed in memory only; never stored or sent off-site
- No facial recognition, no worker identification
- Logs hold timestamp, camera ID and violation type; snapshots are optional and blurred
- Retention period is configurable

Known limits

- Not a certified functional-safety device; never the sole safeguard
- Occlusion, backlight, dust and night scenes reduce accuracy
- Multi-frame voting and confidence thresholds cut false alarms; critical events can get human review

Stage II Targets and How We Will Measure Them

Metric	Target	Method
NPU inference latency (INT8)	≤ 20 ms, p95, incl. USB	HailoRT timing
End-to-end alert latency	≤ 500 ms, p95	Camera event to alert
Concurrent streams	≥ 4 × 1080p @ 15 FPS	Log host CPU load
Accuracy (INT8)	mAP@0.5 ≥ 0.85; drop ≤ 2 pts	Held-out test set
Alert quality	Recall ≥ 90%; ≤ 1 false alarm per camera-hour	Annotated test footage
Accelerator power	≤ 3 W average	USB power meter
Stability and privacy	72 h run; 0 frames off-site	Soak test, packet capture

These are planning goals; measured results will be published in the repository.

Expected Benefits and Business Potential

Target customers

SME factories, construction subcontractors and warehouses that already have IP cameras and an on-site PC.

Customer value

Add a USB accelerator instead of replacing hosts. No cloud subscription, and local-only processing eases privacy talks with workers.

Business model

One-time hardware plus a per-site or per-camera software license. Stage II will quantify cost per camera against cloud analytics.

Beyond PPE: forklift-proximity and posture alerts on the same pipeline; optional on-device summaries of safety events (stretch goal).

Stage II Plan, Timeline and Deliverables

Period	Milestone
Nov 2 – 8	Dataset and FP32 baseline; set up the Dataflow Compiler on a PC
Nov 9 – 15	.hef compilation, INT8 calibration, first UGen300 benchmarks
Nov 16 – 22	HailoRT multi-stream pipeline, rule engine, alert output
Nov 23 – 29	FastAPI dashboard, full benchmarks, 72 h soak test
Nov 30 – Dec 4	Demo video, README, architecture diagram, submission

.hef model, Python HailoRT application, local dashboard and benchmark report, all in the GitHub repository.

References and Repository

Platform and tools

- ASUS UGen AI League rules and UGen300 platform page
- Hailo Model Zoo, HailoRT and Dataflow Compiler documentation
- Ultralytics YOLO documentation (ONNX export, AGPL-3.0)

Datasets

- Roboflow Universe PPE detection datasets
- SH17 PPE dataset for manufacturing
- Open Images V7 (person images)
- Licenses are verified before training

GitHub: https://github.com/beatz19960/2026ASUS-UGEN-AI-LEAGUE-